

Data Communications

Lecture #6

What we have discussed

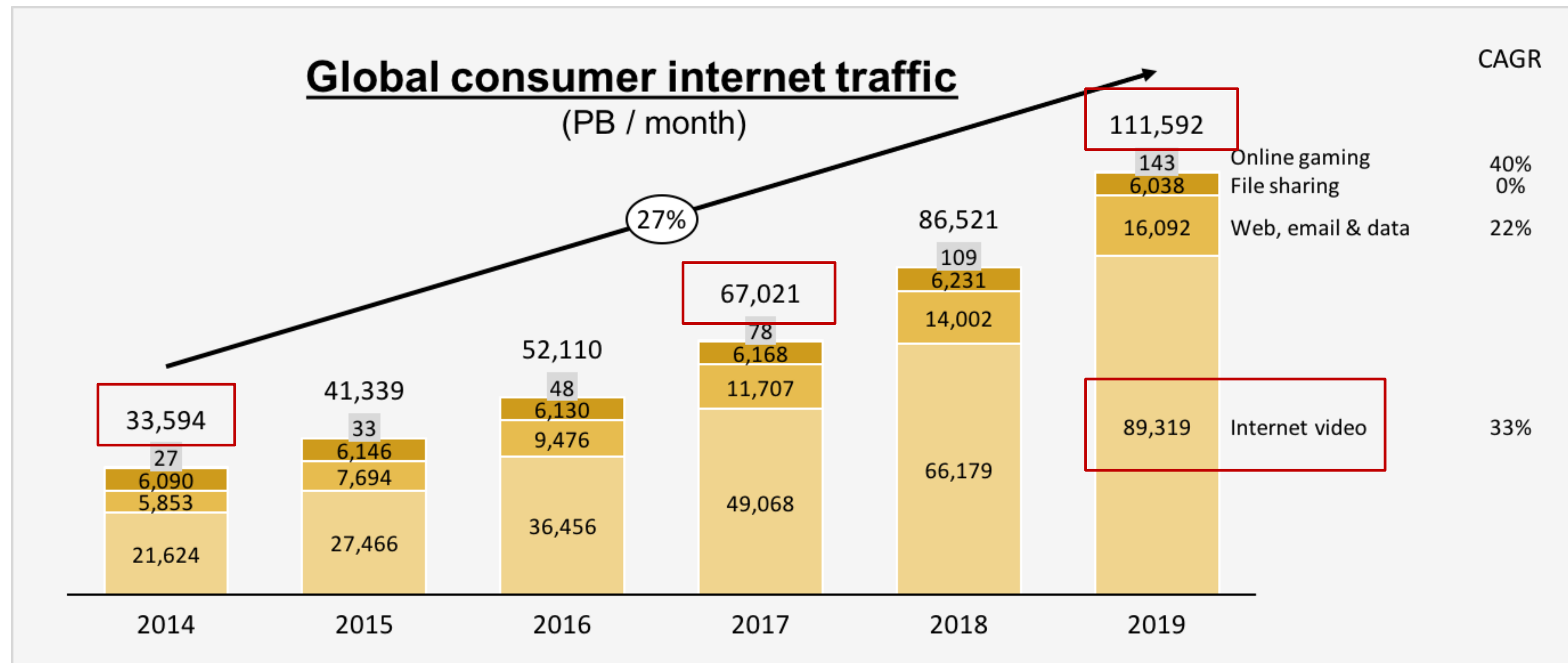
- P2P applications - bitTorrent, DHT

Today...

- Video Streaming / Content Distribution Networks (CDN)

Video streaming and CDN

- Stream video traffic: major consumer of Internet bandwidth
 - Netflix, YouTube, Amazon Prime: 80% of residential ISP traffic (2020)



YouTube

NETFLIX

Akamai

迅雷看看
www.kankan.com
网络高清影院

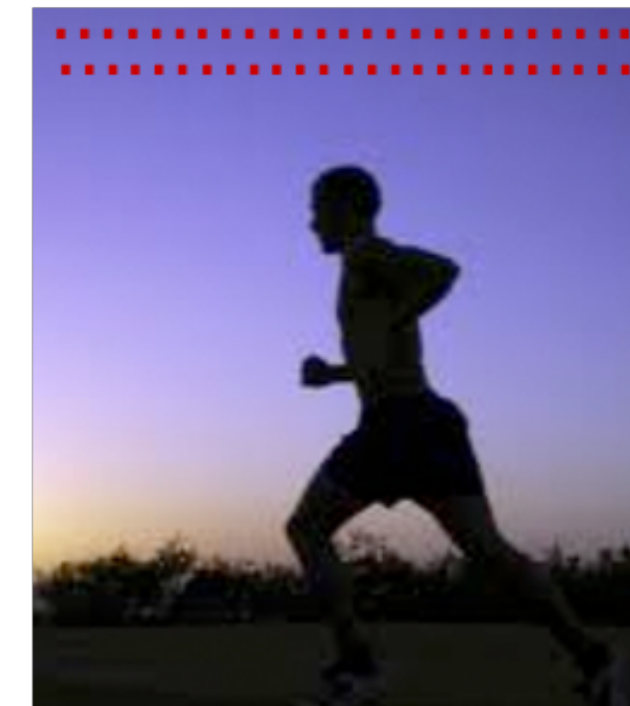
Video streaming and CDN

- Challenge: scale!
 - How to serve ~1B users?
⇒ Solution: distributed infrastructure
- Challenge: heterogeneity
 - Different users have different capabilities (e.g., wired vs. mobile; bandwidth rich vs. bandwidth poor)
⇒ network-independent (application-level) infrastructure

Multimedia: video

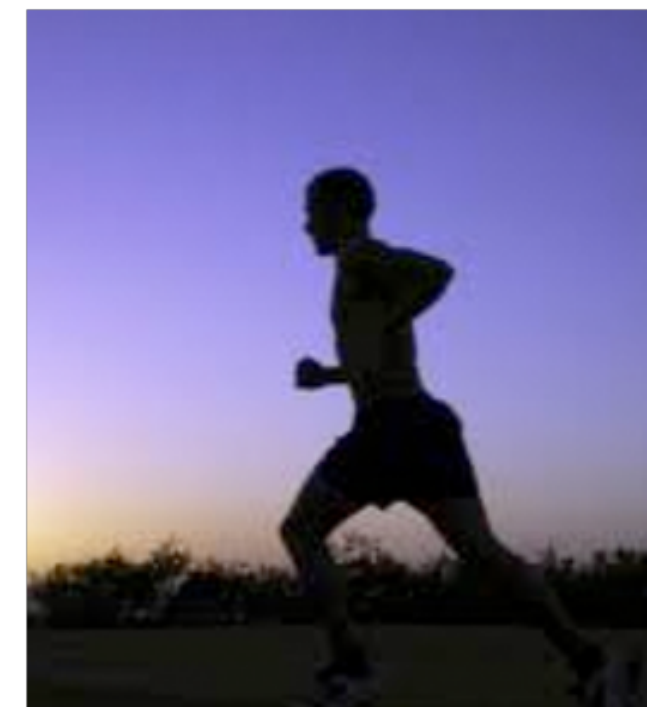
- Video: sequence of images displayed at constant rate
 - e.g., 24 images/sec
- Digital image: array of pixels
 - Each pixel is represented by bits
- Compression (coding): use redundancy within and between images to decrease # of bits used to encode images
 - Spatial (within image)
 - Temporal (from one image to next)

spatial coding example: instead of sending N values of same color (all purple), send only two values: color value (*purple*) and number of repeated values (N)



frame i

temporal coding example: instead of sending complete frame at $i+1$, send only differences from frame i

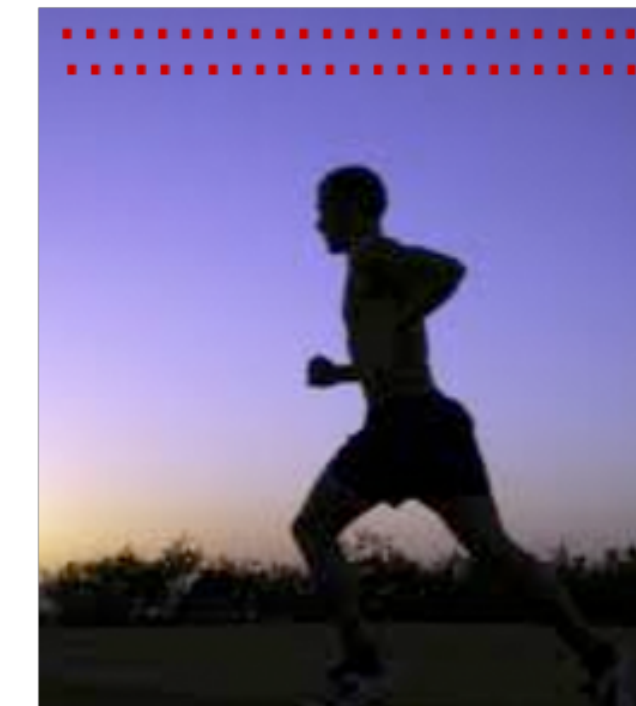


frame $i+1$

Multimedia: video

- Constant Bit Rate (CBR)
 - Video encoding rate fixed
- Variable Bit Rate (VBR)
 - Video encoding rate changes as the amount of spatial, temporal coding changes
 - e.g.,
 - MPEG1 (CD-ROM): 1.5Mbps
 - MPEG2 (DVD): 3-6Mbps
 - MPEG4 (often used in Internet, 64Kbps ~ 12Mbps)

spatial coding example: instead of sending N values of same color (all purple), send only two values: color value (*purple*) and number of repeated values (N)



frame i

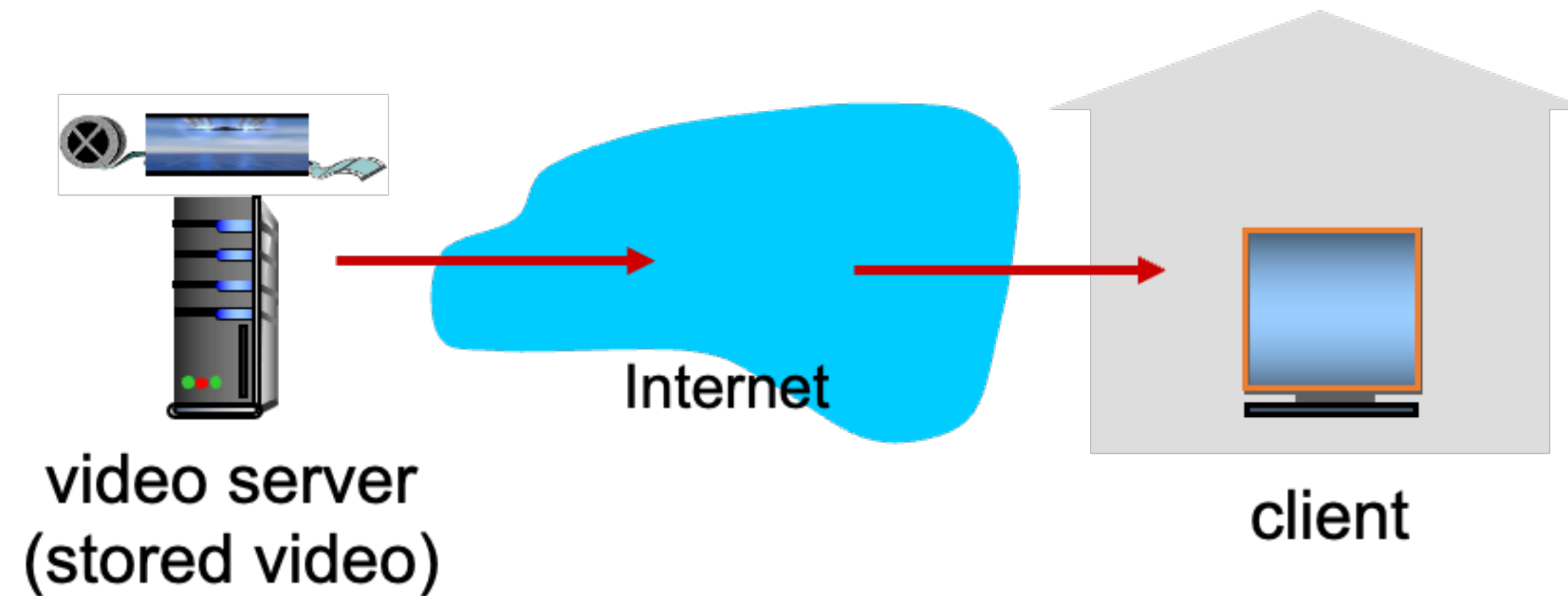
temporal coding example: instead of sending complete frame at $i+1$, send only differences from frame i



frame $i+1$

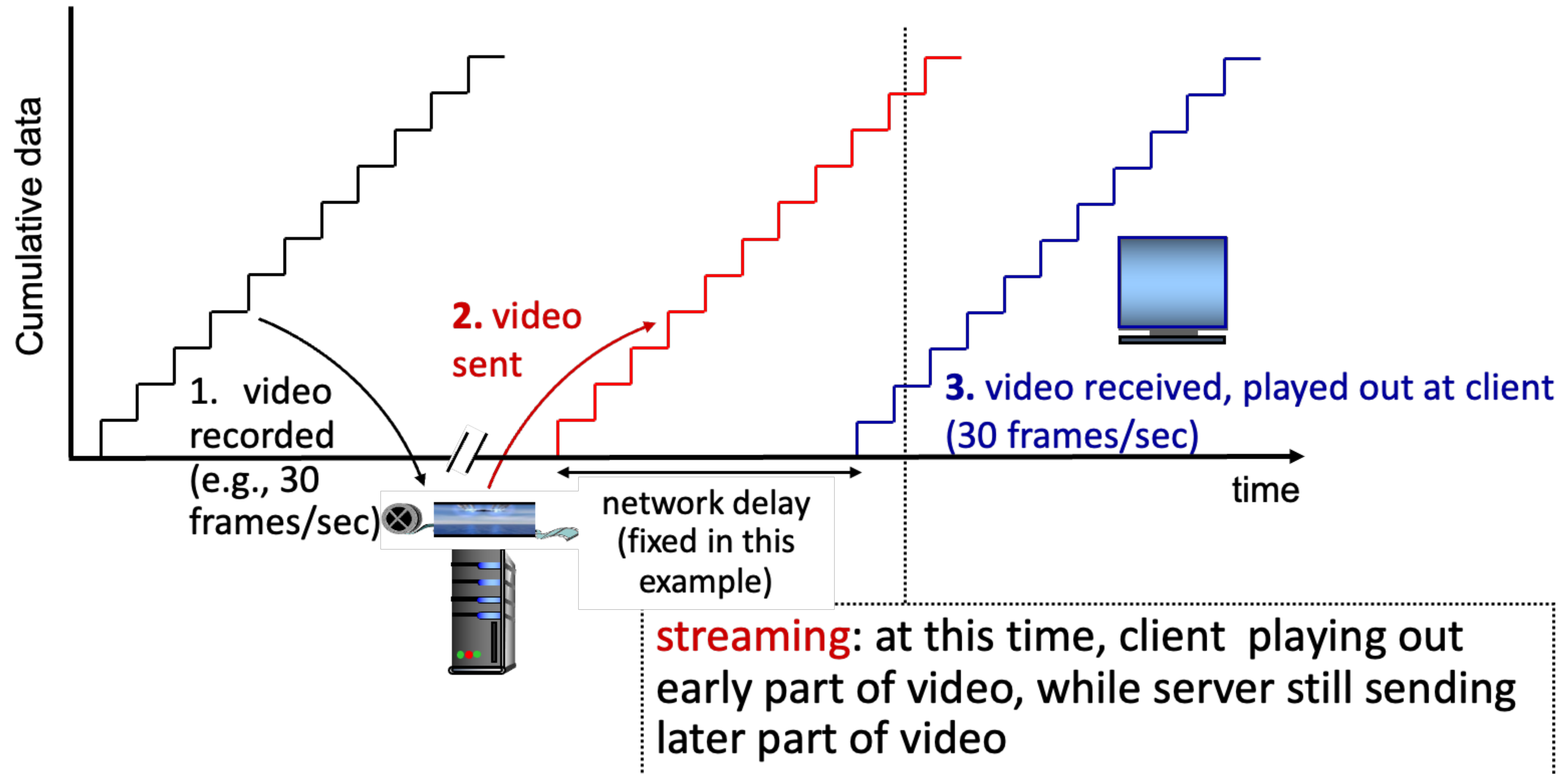
Streaming stored video

- Simple scenario



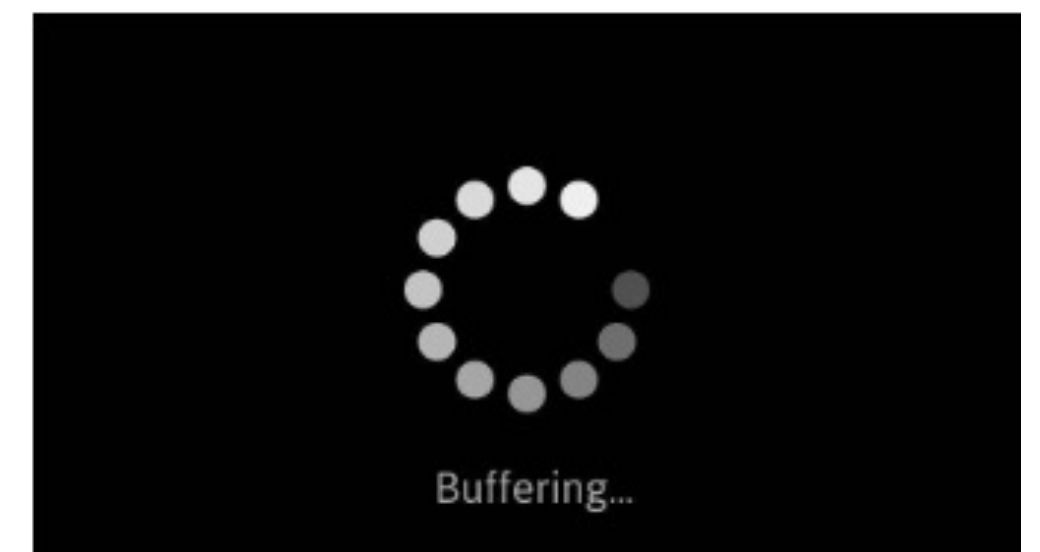
- Challenges
 - Server-to-client bandwidth varies over time due to changing network congestions levels
 - Packet loss and delay due to congestion will delay payout, or result in poor video quality

Streaming stored video

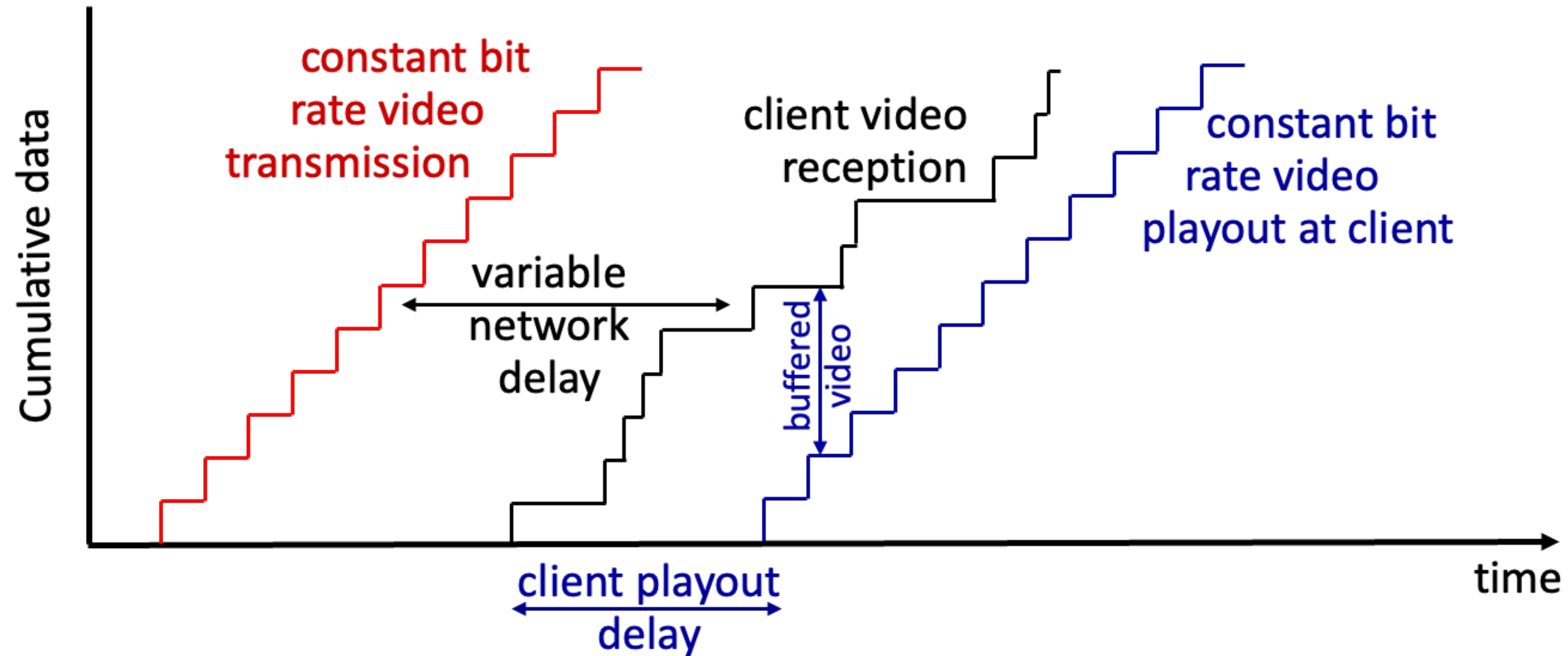


Streaming stored video

- Continuous layout constraint
 - During client video playout, playout timing much match original timing
 - ... but network delays are variable (jitter), so will need client-side buffer to match layout requirement
- Other challenges
 - Client interactivity: pause, fast-forward, rewind, jump through video
 - Video packets may be lost, retransmitted



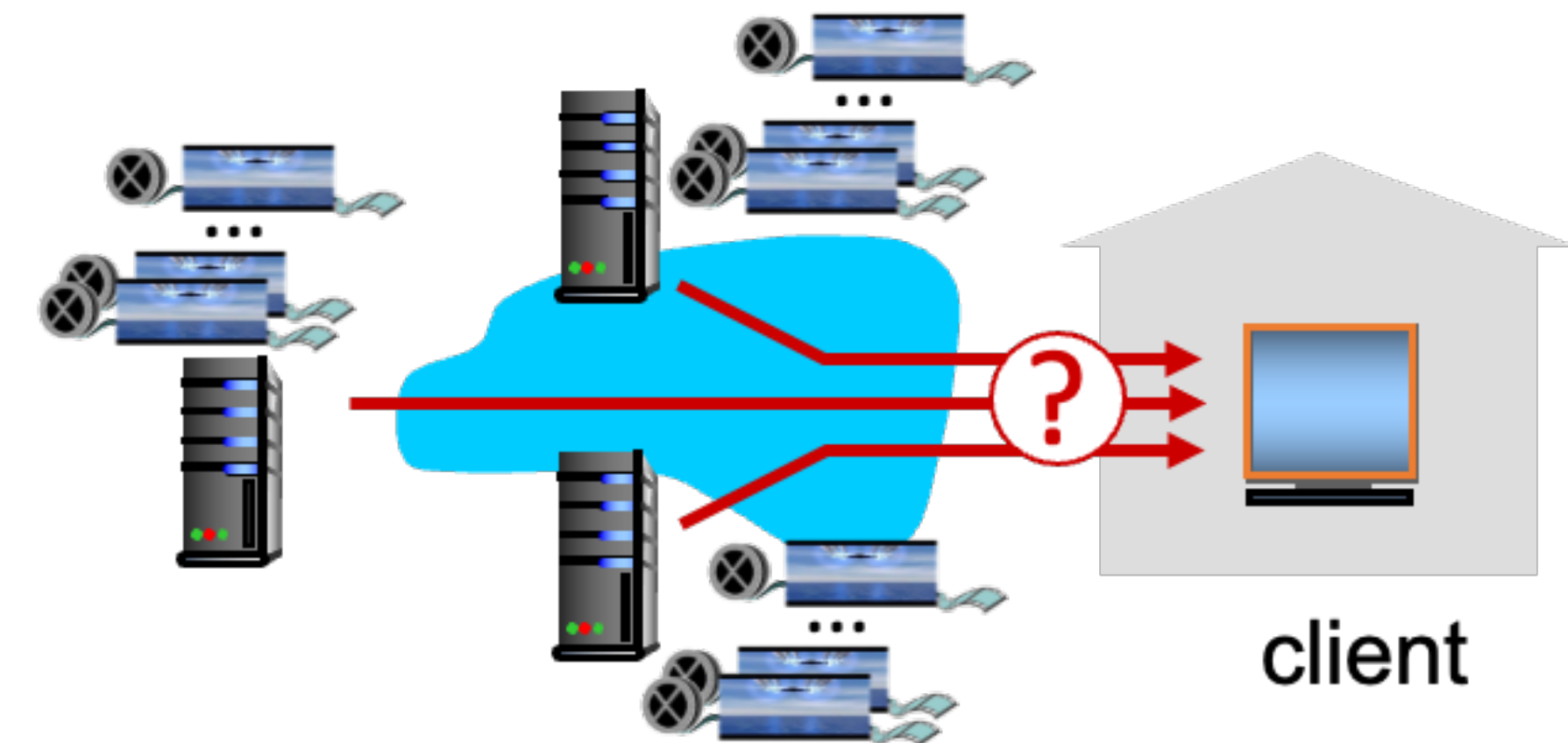
Streaming stored video: playout buffering



- Client-side buffering and playout delay: compensate for network-added delay and delay jitter

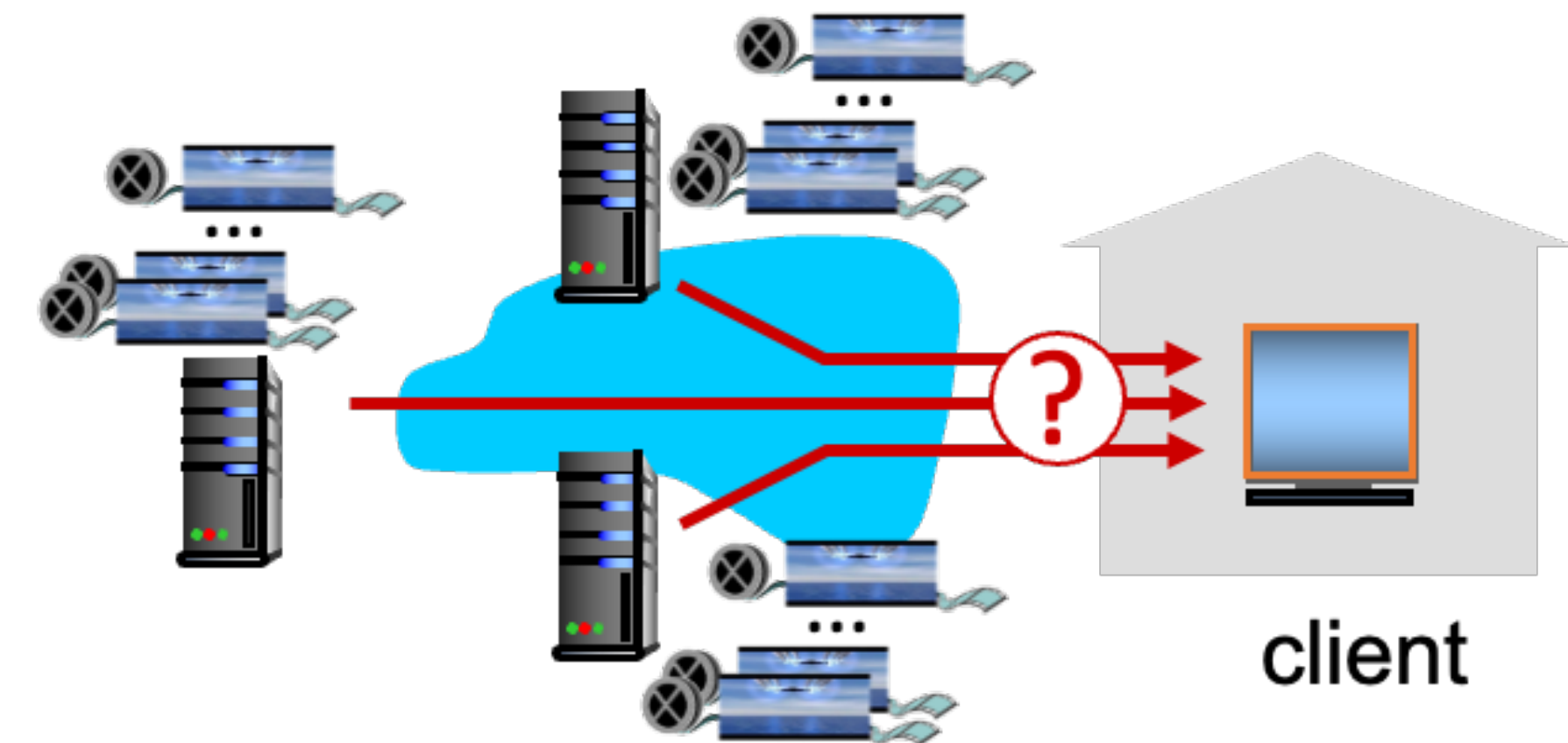
Streaming multimedia: DASH

- DASH: Dynamic, Adaptive Streaming over HTTP
- Server
 - Divides video file into multiple chunks
 - Each chunk is encoded at multiple different rates
 - Each version has a different quality level
 - Different rate encodings are stored in different files
 - Files are replicated in various CDN nodes
 - **Manifest file**: provides URLs for different chunks



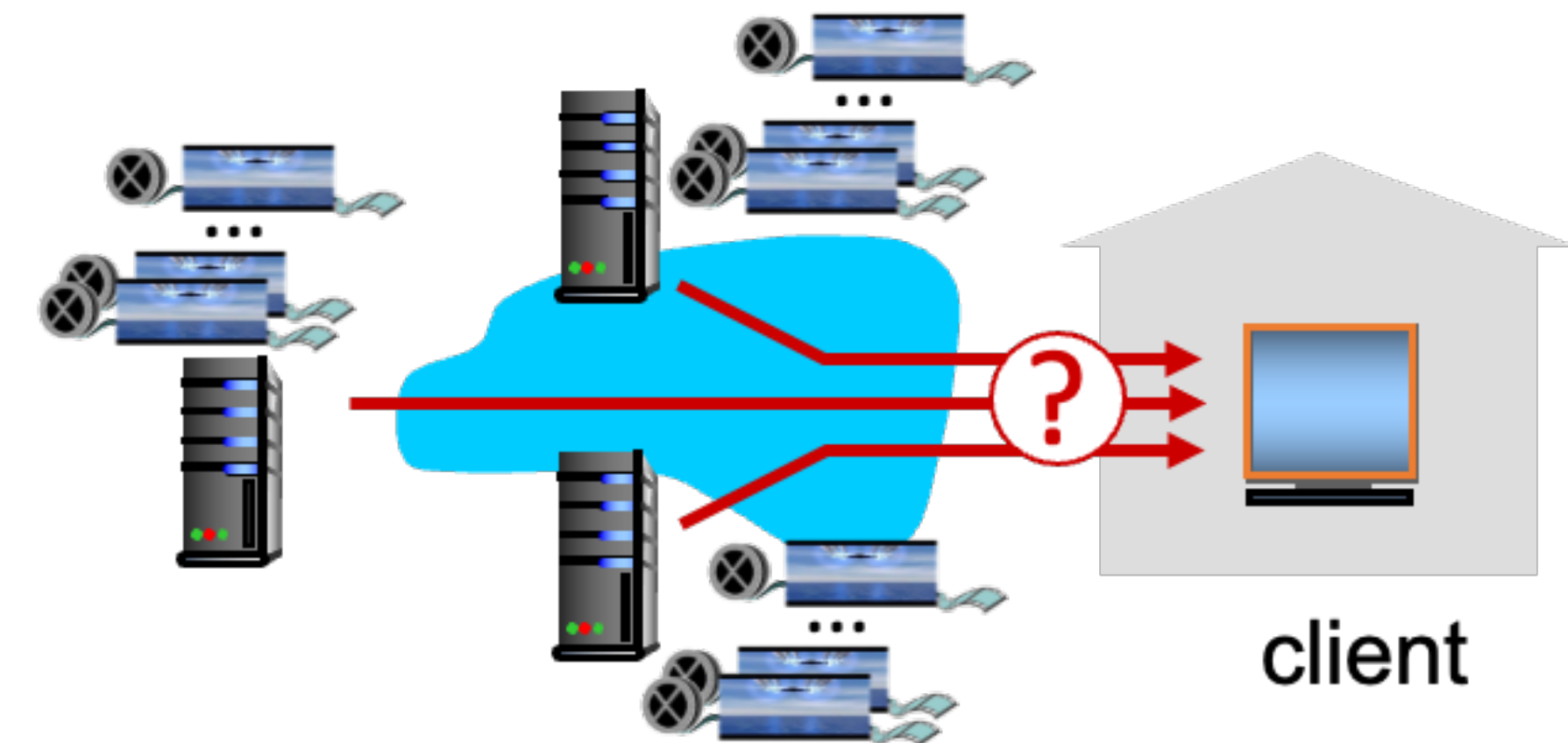
Streaming multimedia: DASH

- DASH: Dynamic, Adaptive Streaming over HTTP
- Client
 - Periodically estimates server-to-client bandwidth
 - Consulting manifest, requests one chunk at a time
 - Choose maximum coding rate sustainable given current bandwidth
 - Can choose different coding rates at different points in time (depending on available bandwidth at time), and from different servers



Streaming multimedia: DASH

- DASH: Dynamic, Adaptive Streaming over HTTP
- “Intelligence” at client: client determines
 - When to request chunks
→ so that buffer starvation, or overflow does not occur
 - What encoding rate to request
→ higher quality when more bandwidth available
 - Where to request chunks
→ can request from URL server that is close to client or has higher available bandwidth



Streaming video = encoding + DASH + playout buffering

Content Distribution Networks (CDNs)

- Challenge: how to stream content (selected from million of videos) to hundreds of thousands of simultaneous users?

option 1) single, large mega server

✓ single point of failure

✓ Point of network congestion

✓ long (and possibly congested) paths to distant clients

⇒ ... quite simply: this solution does not scale

Content Distribution Networks (CDNs)

- Challenge: how to stream content (selected from million of videos) to hundreds of thousands of simultaneous users?

option 2) store/serve multiple copies of videos at multiple geographically distributed sites (CDN)

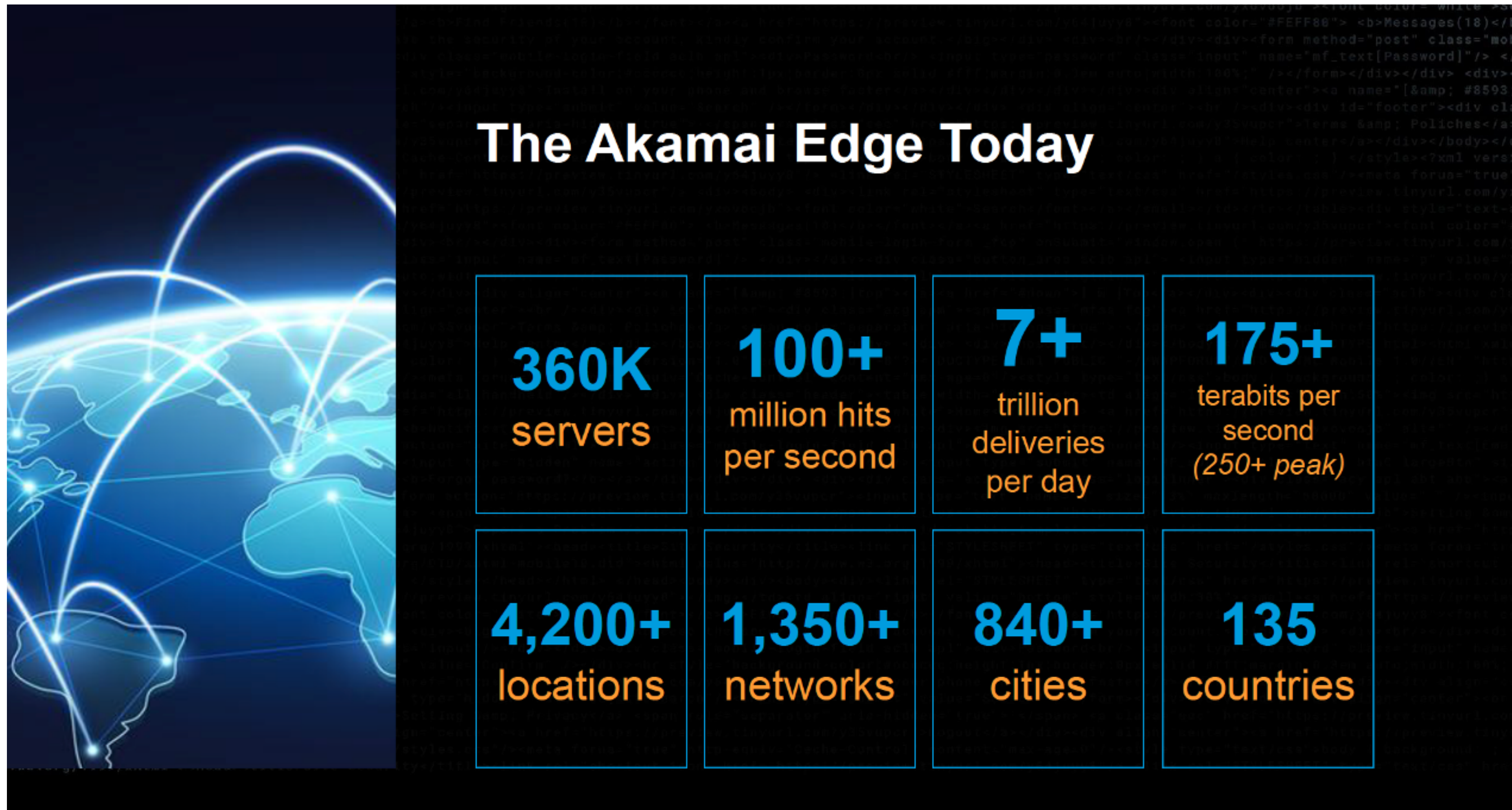
- ✓ Enter deep: push CDN servers deep into many access networks
 - close to users
 - Akamai: 240,000 servers deployed more than 120 countries (2015)



- ✓ Bring home: place smaller number (10's) of large cluster in the point of presences near access network
 - used by Limelight



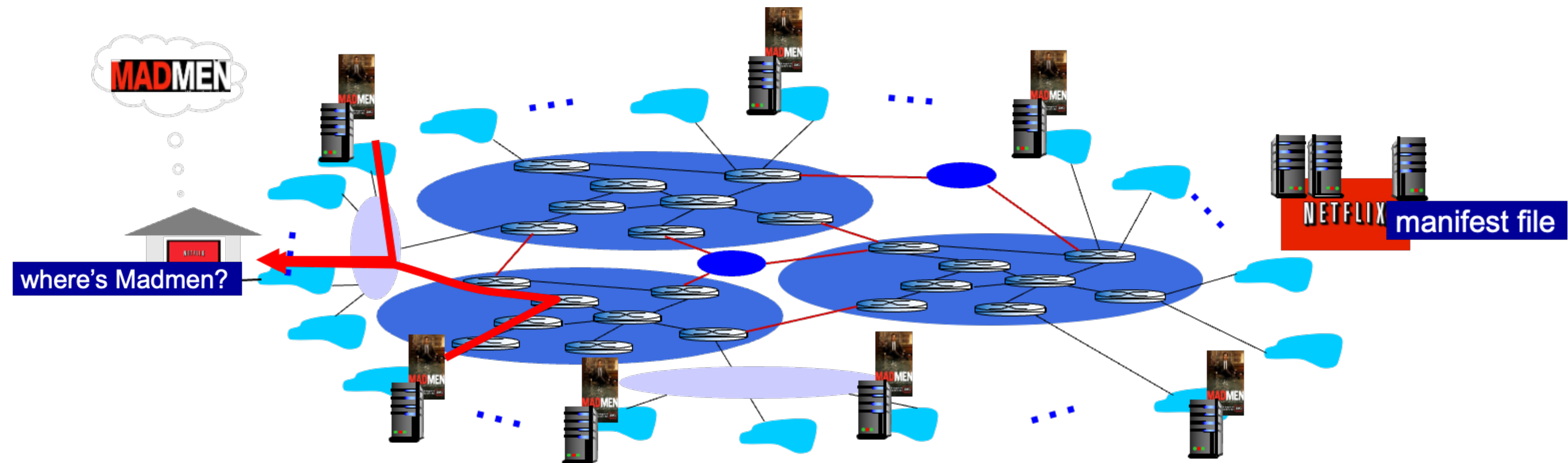
Akamai today



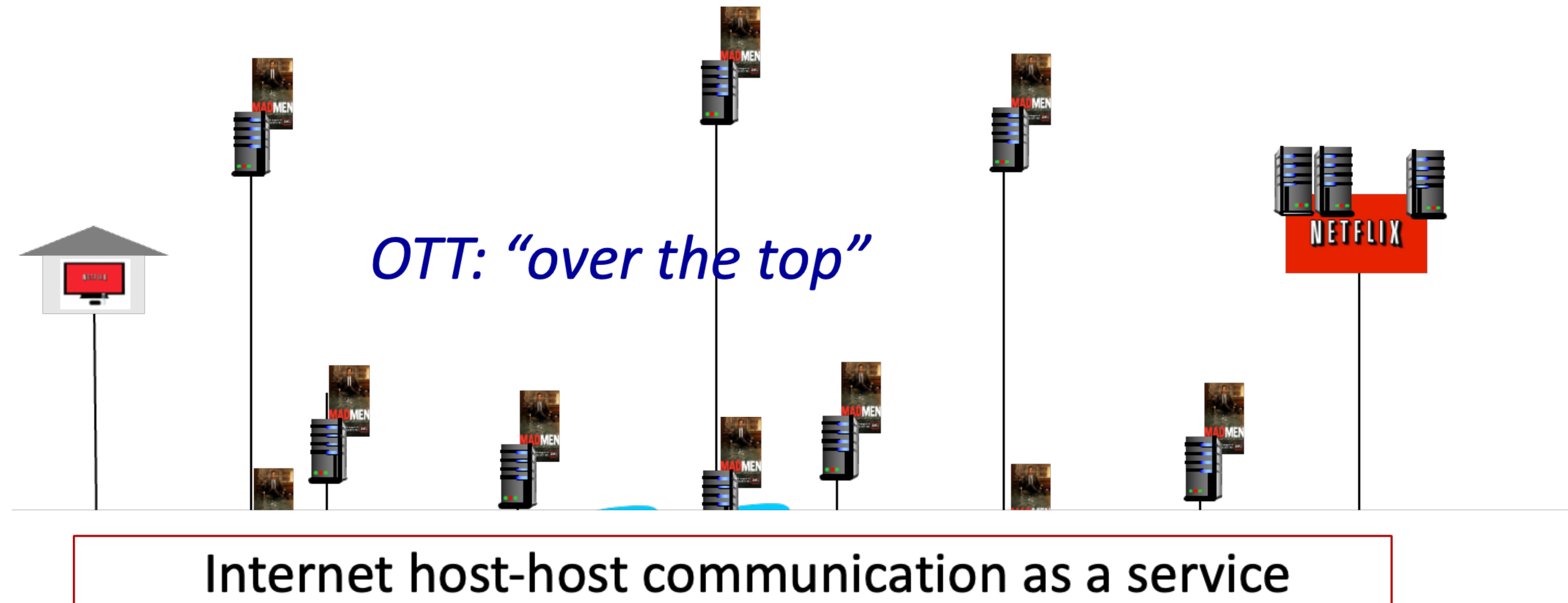
Source: <https://networkingchannel.eu/living-on-the-edge-for-a-quarter-century-an-akamai-retrospective-downloads/>

How does Netflix work?

- Netflix: stores copies of content (e.g., MADMAN) at its worldwide OpenConnect CDN nodes
- Subscriber requests content, and service provider returns manifest
 - Using manifest, client retrieves content at highest supportable rate
 - May choose different rate or copy if network path becomes congested



Content Distribution Networks (CDNs)

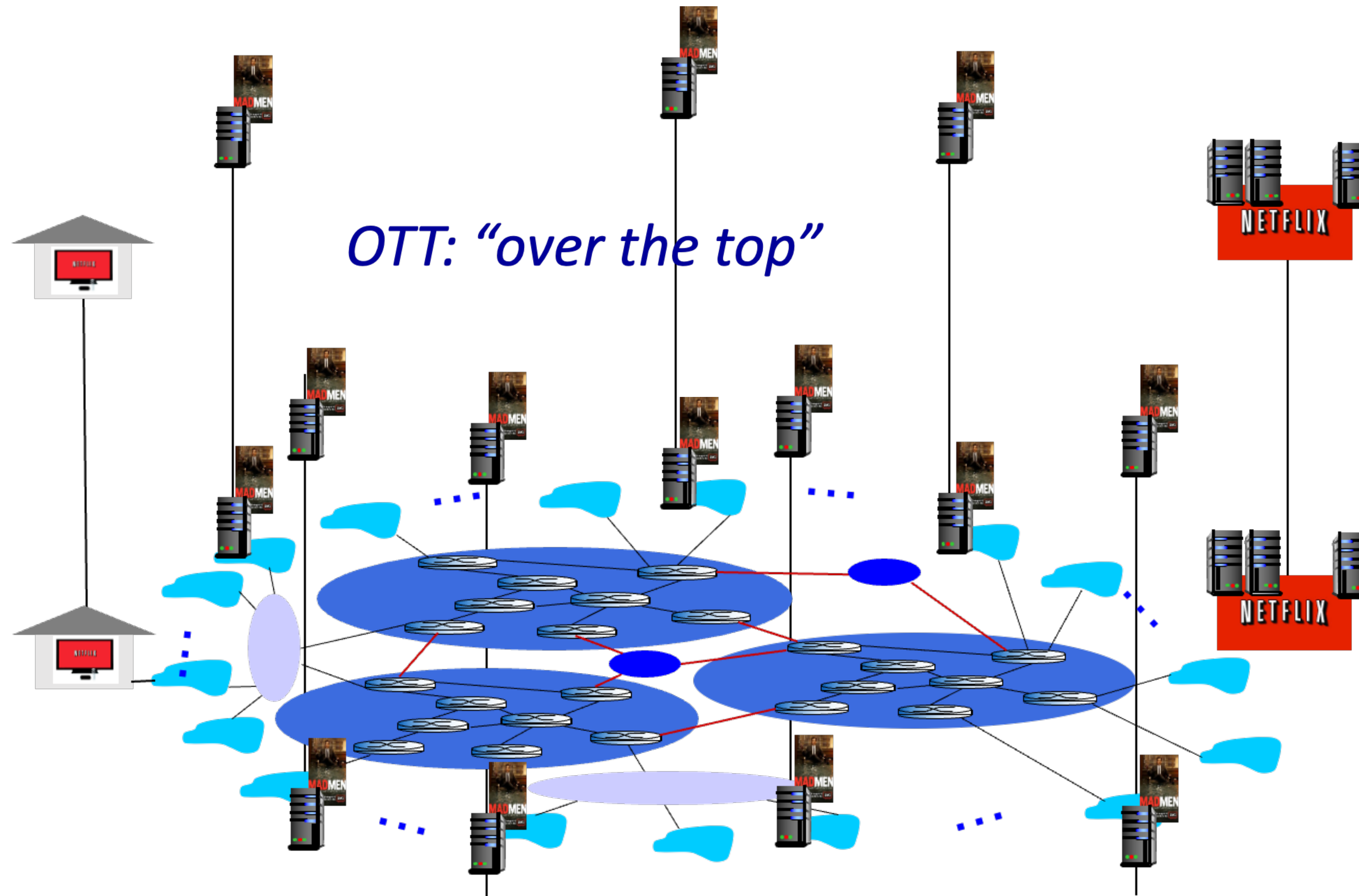


OTT challenges: copying with a congested Internet from the “edge”

✓ What content to place in which CDN node?

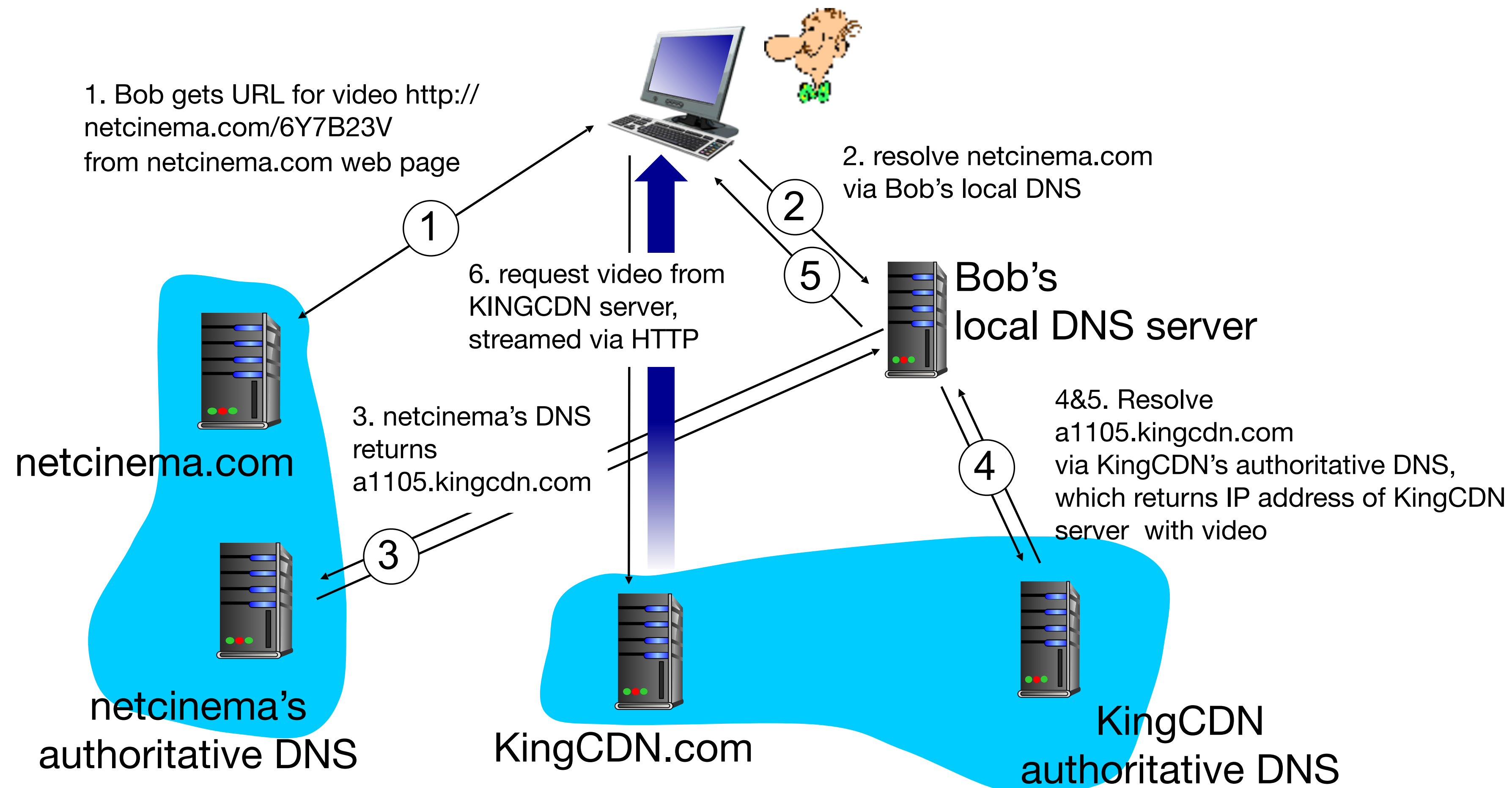
✓ From which CDN node to retrieve content? At which rate?

Content Distribution Networks (CDNs)

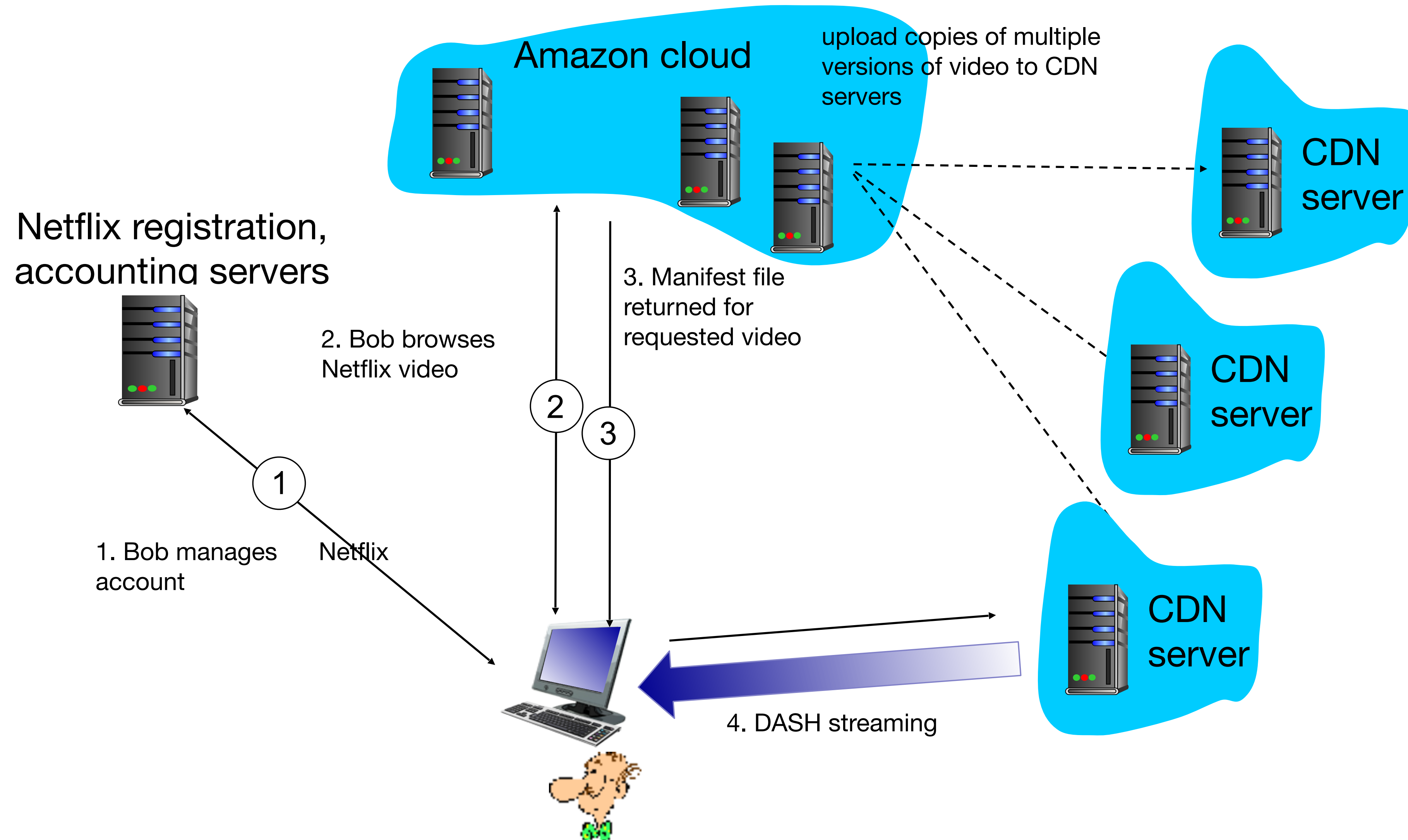


CDN content access: a closer look

- Bob (client) requests video <http://netcinema.com/6Y7B23V> which is stored in CDN at <http://KingCDN.com/NetC6y&B23V>



Case study: NETFLIX



Questions?